SK HUMAIN AHMED

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Mechanical engineer with 8 years of experience in utility-scale computational modeling, digital twins, and field data systems for a ~2-million-meter power distribution network. Highly motivated to pursue a Ph.D. focusing on intelligent dynamical systems, robotics, and physics-informed AI, learning-based control, with applications to energy and cyber-physical systems.

RESEARCH INTERESTS

Data-Driven Modeling & Physics-Informed Machine Learning • Sensing, Perception & Autonomy
Nonlinear Dynamics, System Identification & Intelligent Control • Energy and Smart Grid Management & Control

EDUCATION

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh

B.Sc. in Mechanical Engineering | CGPA: 3.47/4.00 (Senior Year Trend: L4/T1: 3.71, L4/T2: 3.96)

- Standardized Tests: GRE: 311 (V: 151, Q: 160, AWA: 3.5) | TOEFL iBT: 97 (R: 25, L: 24, S: 27, W: 21)

RESEARCH & INDEPENDENT PROJECTS

SINDy-RL for interpretable and efficient model-based reinforcement learning (independent project)

- Reimplemented the SINDy-RL framework (Zolman et al., 2025) on the CartPole swing-up task, using sparse system identification to learn a surrogate dynamics model for policy training.
- Developed a 41-term quadratic control-affine library to identify discrete-time nonlinear dynamics using ensemble-based sparse regression and Sequential Thresholded Ridge Regression (STLRidge) via PySINDy.
- Integrated Proximal Policy Optimization (PPO) with learned SINDy surrogate dynamics using Ray RLlib, employing a neural-network policy with two 64-unit hidden layers with iterative learning pipeline combining offline data collection, surrogate policy training, and online dataset updates to continuously improve surrogate-model accuracy and system control adaptation

Study and Simulation of Various Heat Transfer Models of SI Engine (Undergraduate Thesis)

- Simulated combustion and heat transfer processes of a spark ignition engine using MATLAB.
- Implemented, compared, and evaluated the predictive performance of various heat transfer models, integrating experimental data to determine the most accurate correlations for engine behavior.

PUBLICATIONS & PRESENTATIONS

- **Sk Humain Ahmed**, et al. (2026). "Hybrid Machine Learning Framework for Predictive Maintenance in Smart Manufacturing Using Sensors." Presented at the 9th European International Conference on Industrial Engineering and Operations Management (IEOM 2026), Barcelona, Spain, July 2026.
- Md Shariful Islam, **Sk Humain Ahmed** and Afsar Ahammed (2016). "Study and Simulation of Various Heat Transfer Models of SI Engine." Proceedings of the International Conference on Mechanical, Industrial and Energy Engineering (ICMIEE), Khulna, Bangladesh, 2016.

PROFESSIONAL EXPERIENCE

Northern Electricity Supply PLC Rajshahi, Bangladesh | 2018 – Present

Executive Engineer

- Contributed to the development of a digital twin for a large-scale electrical distribution with ~2 million customer meters, ~20,000 km of lines, and ~13,000 transformers.
- Performed load-flow simulations, short circuit current evaluations, and asset reliability analyses using computational tools to optimize network operational decision-making.
- Independently designed and deployed a production-grade Web application for visualizing electrical distribution network assets / Mobile application for surveying customer meters with complete power connectivity from source to meter point using Python (Django REST Framework), PostgreSQL/PostGIS, and Flutter/Dart integrating offline field sensing, spatial synchronization, and utility-scale asset management.

Eastern Refinery Ltd 2017 – 2018

Probationary Engineer (Mechanical) | Chattogram, Bangladesh

- Conducted mechanical inspection, troubleshooting, and maintenance of complex thermo-fluid process systems, including centrifugal pumps, gas compressors, and steam-turbine units.

Uttara University 2016 – 2017

Lecturer (Part-time) | Dhaka, Bangladesh

- Delivered undergraduate lectures on foundational mechanical engineering and physics principles.

TECHNICAL SKILLS

- Programming & ML: Python, C/C++, PyTorch, PySINDy, Ray RLlib, DeepMind Control Suite, Scikit-Learn
- Engineering & Simulation: MATLAB/Simulink, ANSYS, SolidWorks, AutoCAD, WindMil (Milsoft)
- Software & Databases: PostgreSQL, PostGIS, Django REST Framework, Flutter

CERTIFICATIONS & TRAINING

- Technical Exposure Program on Smart Grid Systems and SCADA/ADMS, NRECA Headquarters, USA (2024)
- Advanced GIS-Based Management System Training – Spatial Database Design, Electrical Network Modeling for Grid optimization, NRECA International (2023-2024)